

The Endocrine System

Below are some notes in orange to review themes and intended take-aways from lab!

This week we are introducing the Endocrine system, including structure and function and microanatomy of the major endocrine organs. You will need to know the hormones that they secrete as well as study their microanatomy.

The endocrine system extends from brain to reproductive organs and has many glands that secrete hormones into the bloodstream. These chemicals are transported throughout the body in blood and bind to target cells that have cell membrane receptors for a specific hormone. Hormones cause changes in activity in these target cells (which may be another endocrine gland, or another type of cell/organ) and direct a variety of cellular activities to keep the body in homeostasis.

Materials:

Endocrine Slides (see below):

Ovary

Pituitary

Adrenal Gland

Thyroid

Hypothalamus

Testis

Microscope Accessory Boxes

Stopwatches

Stop Boxes

Automatic Blood Pressure Cuffs

Digital Metronomes

Bucket of Ice/water

Set of Math Problems

Graph Paper

Colored Pencils

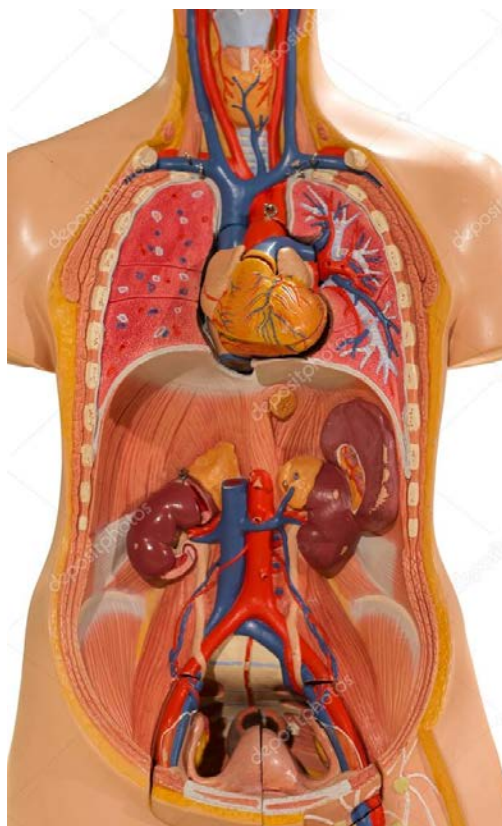
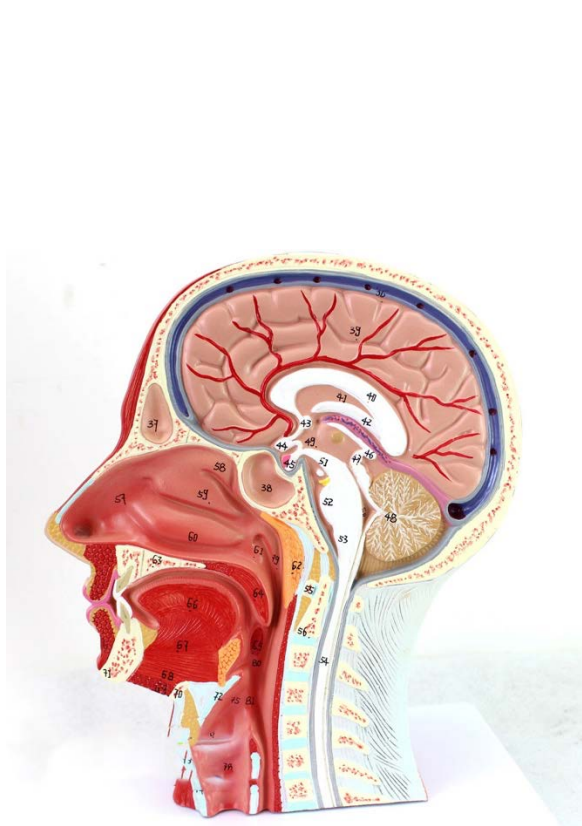
Empty bottles for endocrine related pharmaceuticals

ACTIVITY ONE – FIND THE ANATOMY ON A LAB MODEL

With one lab buddy – go find on a model (you'll need to use more than one model to find all of these) in lab these endocrine organs/tissues. Check them off as you find them:

- ☐ pineal gland
- ☐ hypothalamus
- ☐ pituitary gland
- ☐ thyroid gland
- ☐ parathyroid glands
- ☐ adrenal glands
- ☐ pancreas
- ☐ ovaries and testes

Be able to identify the structures above on a diagram or model (some pictures of relevant models – similar to the ones we have in lab – are below).



Activity 2 – Notebook Questions

Answer these questions in your lab book/notebook. Feel free to draw/add color to anywhere that helps!

The answers to the questions below can simply be looked up online or in the book. Don't feel the need to go into much detail! Sometimes a single word is enough and other times a short phrase/short sentence is sufficient.

- a. Write out the main differences between an ENDOCRINE and an EXOCRINE gland
- b. Explain how the pancreas has BOTH exocrine and endocrine parts/functions
- c. Explain the relationship between the hypothalamus and the pituitary gland using suppressing and releasing hormones
- d. What are the two hormones secreted by the **posterior pituitary**?
 - a. What are the cells called that secrete these two hormones?
 - b. Where are the cell bodies of these cells located?
- e. The **anterior pituitary** secretes several different types of hormones.
 - a. What hormones are secreted here?
- f. What is the histological structural unit of the thyroid gland called?
- g. What are the names of the two major thyroid hormones?
 - a. What is the name of the cells that secrete these thyroid hormones?
 - b. What is the name of the cells that secrete calcitonin?

- h. What is the name of the cells that secrete parathyroid hormone in the parathyroid gland?
- i. ADRENAL GLAND:
 - a. Below the capsule, what is the name of the outermost layer of the adrenal cortex and what hormone does it secrete?
 - b. What is the name of the middle layer of the adrenal cortex and what hormone(s) does it secrete?
 - c. What is the name of the inner layer of the adrenal cortex and what hormone(s) does it secrete?
 - d. What is the name of the cells in the adrenal medulla and what hormones do they secrete?
- j. Describe the hormones that come from the Ovaries and Testes in the Female/Male reproductive organs

Activity 3: Microscopic anatomy of Endocrine Glands

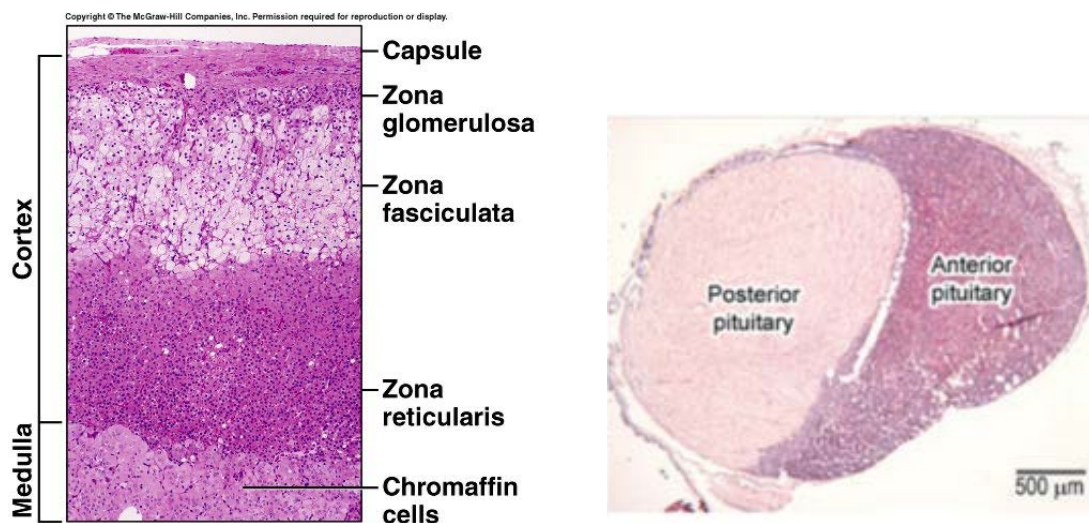
Your instructor has laid out several slides. Examine the histology of glands indicated and for each:

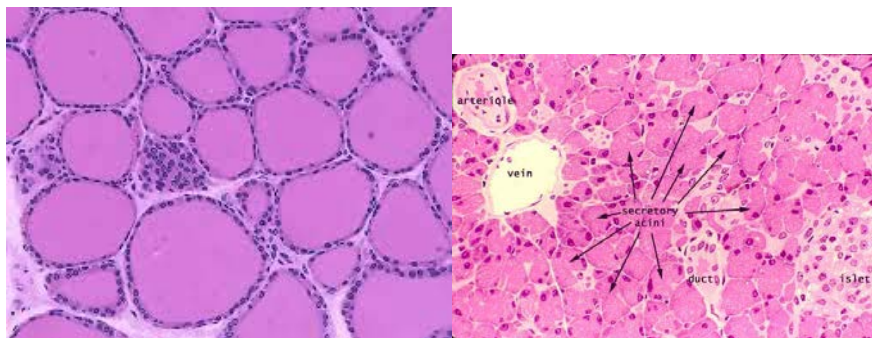
1. Be able to recognize each organ from histological sections.
2. Identify major endocrine structures or cells (e.g., Islets of Langerhans)
3. Know hormones produced by tissue and their effects

Based on the slides you looked at in lab, be able to recognize

- Anterior vs posterior pituitary gland
- Thyroid gland tissue
- Pancreas tissue
- Adrenal gland tissue, including the zona glomerulosa, zona fasciculata, zona reticularis, and medulla

Below are some examples of these microscope views (not in the same order as listed above!)





Activity 4: Role of A.N.S. and Endocrine System in Stress Response

Purpose: To understand the physiological basis of the Stress Response

Directions:

1. Within your group choose TWO of the four stress activities listed below (Labeled A-D). Every person in your group will perform those same 2 activities, so you will generate averages within your table group (won't be just one person at your table doing this!).
2. Research Question: What effect will each stressor have on heart rate (HR) and blood pressure (BP)?
3. Form a hypothesis for each stressor tested (e.g., "Pain will elevate HR")
4. Make measurements for every person at your table for each of the stress activities as directed below and record data in the provided tables.
5. Calculate average values for HR & BP at each time point for the data collected at your table
6. Construct bar graphs depicting HR & BP *before*, *during*, and *after* the stress is applied (be sure to average group data- don't draw a bar for each person)
7. In the space below, outline a set of nervous system responses that could account for the changes in HR and BP that you see across your graphs.

Consider the way the sympathetic nervous system might initiate these changes.

8. In the space below, outline a set of hormonal responses that could account for the changes in HR and BP that you see across your graphs.

Consider the way the release of epinephrine and norepinephrine (and perhaps other stress hormones) might initiate these changes.

9. Why do you think it takes time (often a couple minutes or more) for your HR and BP to recover from a stressor? That is, why don't those values more quickly return to baseline values when a stressor is removed?

Consider the fact that, unlike nervous system signals that last only a very short time, hormones can circulate for quite a while and continue to have impacts on the body.

Stress Activities (Choose 2 - every person in your team will complete those same 2)

A. Temperature Stress

1. Sit quietly for 5 minutes.
2. Record resting HR and BP. Deflate cuff but leave attached to one arm.
3. Place hand in a bucket of ice water.
4. Take HR and BP measurements at 1 minute intervals for 3 minutes.
5. At 3 minutes, remove hand from the ice water and pat dry.
6. Record post-stress HR and BP 2 min after removal.

B. Orthostatic Hypotension

1. Sit quietly for 5 minutes.
2. Record resting HR and BP. Deflate cuff but leave attached to one arm.
3. Stand upright against a wall. It is important to relax as much as possible.
4. Record HR and BP immediately upon standing.
5. Measure HR and BP again 2 min and 4 min after standing. Stop the experiment if you begin sweating or feel light-headed.
6. At the end of the time period, sit down and rest. Record post-stress HR and BP 2 min later.

C. Exercise Stress

1. Sit quietly for 5 minutes.
2. Record resting HR and BP. Deflate cuff but leave attached to one arm.
3. Set metronome at 98 beats/ minute. Step up and down on the step-box, keeping pace with the metronome (approximately 24 steps/ min).
4. Record HR and BP at 1 minute intervals for 3 minutes. It will be necessary for a student to support the sphygmomanometer during the exercise. The experiment should be terminated if you feel faint or out of breath.
5. Stop exercise at three minutes and sit and rest.
6. Record post-stress HR and BP 2 minutes after end of exercise.

D. Academic Stress

1. Sit quietly for 5 minutes.
2. Record resting HR and BP. Deflate cuff but leave attached to one arm.
3. Complete sheet of math problems as quickly as possible (maximum time allowed = 3 minutes).
4. Record HR and BP at 1 minute intervals.
5. Record post-stress HR and BP 2 minutes after end of treatment.

Activity 5: Pharmaceuticals and the Endocrine System



The endocrine system is one of the systems most commonly targeted by pharmaceutical drugs. Numerous things about hormones make them excellent drug candidates: They circulate to almost every cell, often remain in the body for hours, and can have major impacts across the entire body or targeted impacts on only certain specific cells, depending on the locations of their receptors. Some endocrine related drugs are exact copies of naturally occurring human hormones (hormone replacement therapies) while others are chemicals not normally found in the body but aimed at stimulating or inhibiting endocrine cells in certain glands.

1. Obtain one of the empty bottles for an endocrine related pharmaceutical drug
2. Use web research to find out more about that drug. Record the following:
 - a. The generic name of the active chemical in the drug
 - b. The trade name of the drug
 - c. The main disease or condition for which the drug is used
 - d. If the drug is for hormone replacement (i.e., the drug chemical is a copy of a naturally occurring hormone or is made to mimic a naturally occurring hormone)...
 - i. What endocrine organ normally produces that hormone?
 - ii. What region or cell within that organ normally produces the hormone?
 - iii. What can you find out about why a person's body would not be producing sufficient amounts of the hormone and would therefore need hormone replacement?
 - e. If the drug chemical is not meant to take the place of a naturally occurring hormone, but rather is meant to act on the endocrine system to stimulate or inhibit its activities...
 - i. What endocrine organ will be most directly impacted/targeted by this drug?
 - ii. What region or cell within that endocrine organ will be most directly impacted?
 - iii. What will that endocrine organ do as a result of exposure to this drug?

For this section, you should know the following regarding a few of the drugs from class

- The difference between the generic/active chemical name of the drug and the trade name of the drug
- The main health context for using the drug (e.g., what disease or condition it's used for or why someone would take it)
- Whether or not the drug represents hormone replacement (e.g., putting more of a naturally existing hormone into the body)
- What endocrine organ would naturally produce that chemical or would be most directly impacted by the chemical
- What region of the organ or cell within the organ would normally produce the chemical or will be most directly impacted
- How the body will respond to the drug

Below are pictures of the drug examples that were available in class.







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